Response of STF Influenced Lattice Structures Under Complex Loading

Yujin Kim

April 27, 2026

Word Count: 4102

Response of STF Influenced Lattice Structures Under Complex Loading

The World Health Organization emphasizes the importance of protective gear when reducing injury. However, there is an estimate that approximately 2.3 million people that encounter fatal accidents due to work related causes according to the International Labour Organization. Currently, these protective systems rely on metals and fibers that add uncomfortable weight, along with them being expensive (Alexe, 2024). Protective gears such as objects like helmets, bumpers, car crash absorbers, and shields are crucial in fields like civil engineering. These objects need to meet necessary requirements such as being able to withstand wear and tear, thrive under different environmental conditions, and absorb or reflect energy when in sudden contact. This leads to the integration of shear thickening fluids (STFs) as they are a potential solution to this problem. The state of STFs are fluid under normal circumstances because they are suspensions of solid particles in a liquid carrier, but they become extremely stiff under high shear rates. This type of transition occurs at the particle level as a result of frictional interactions, dilation, and clustering, which efficiently stresses the suspension and transforms it into the most solid state possible (Wei, 2022). Because of this behavior, STFs tend to absorb and release lots of energy when there are sudden impacts while still being lightweight and flexible in daily life. Due to this adaptable quality STFs give, they are suitable for protective applications when comfortability is just as needed as protection foams and fabrics that are engulfed in STFs are more resistant to unexpected attacks than the untreated objects (Liu, 2023).

However, there is a clear gap in the literature of research combining STFs with metamaterials in complex, multiaxial loading conditions that are feasible. STF fluids should be reinforced by metamaterials, which are composites that gain its properties from its structure rather than itself. These tests are crucial before fully emerging in appliances as multiaxial loading

simulates the real world. These complex structures can be precisely created through recent developments in additive manufacturing technologies, such as 3D printing lattice structures from thermoplastic polyurethane (TPU) to engulf STFs for testing. Ultimately, this would help test conditions that are feasible on the universal testing machine (UTM) for multiaxial loading.

Because real world protective materials are rarely subjected to stresses in a single direction, this topic is crucial. This research offers a more accurate assessment of STF metamaterial systems by examining biaxial and multi-axial loading. By incorporating varying STF concentrations into 3D printed TPU lattices and evaluating them under compression, shear, torsion, and combined stress conditions, the experiment's design will fill this gap. This experimental structure makes it possible to assess lattice geometries, fluid compositions, and strain rates.

There are no established experimental designs for measuring the connection of STF rheology and the metamaterial geometry in dynamic events, and its interaction. Engineered lattices such as auxetic, honeycomb, and chiral designs all expand horizontally when they are stretched which increases the effectiveness on load redistribution enhancing energy absorption. In contrast, chiral and honeycomb lattices have predictable patterns of deformation that enhance resilience and prevent catastrophic failure. These complex structures can be precisely created through recent developments in additive manufacturing technologies, which makes it easier to produce lightweight materials that function better than original materials (Wei, 2022). However, the majority of research on mechanical metamaterials has concentrated on uniaxial loading conditions. Leading to the research question of: *How can the integration of shear thickening fluids and metamaterial structures enhance energy absorption and mechanical performance under complex loading?*

Hypothesis

This study hypotheses that incorporating shear thickening fluids to 3D printed TPU metamaterial lattices will improve energy absorption, stiffness, and capacity to support loads under complex multiaxial loading. Because of the fluid's shear thickening behavior, the effect would be more noticeable at higher strain rates. It is also anticipated that various lattice geometries will interact with STFs in unique ways, with auxetic structures demonstrating the highest resistance. When STF is added to metamaterial lattices, it is predicted that the experimental results will show a quantifiable improvement in performance metrics of deformation resistance and strain distribution. It is expected that the distinct patterns of lattice geometries under multiaxial loading will show which structure provides the best reinforcement when paired with STF. Furthermore, it is suggested that the findings will determine the ideal range of STF concentrations and lattice parameters that produce the best possible balance between flexibility and impact resistance.

This research intends to develop flexible lattice structures that can be infused with different concentrations of STF by utilizing 3D printers. This will enable controlled research into material interactions and structural mechanics. Establishing a scalable framework for designing protective materials that optimize energy dissipation will ultimately make them more feasible for use in civil engineering systems, safety equipment, and automotive components.

Literature Review

Shear thickening fluids (STFs) are nanosuspensions that are a reversible low-viscosity liquid that can rapidly transition to a solid-like state when the shear rates reach a critical value. This behavior allows for significant viscosity increases of many orders of magnitude (Wei, Lin,

& Sun, 2022). For many years, the rheological behavior and possible applications of shear thickening fluids have been the main focus of research. Immediate change in viscosity caused by particle formation at high shear rates were the subjects of early research. Because of the significant impact energy released by this shear induced hardening, STFs have been integrated into protective gear systems, where they significantly increase energy-absorption capacity and postpone fabric penetration. The goal of current research has been to find applications to expand these fluids in real life other than just protective gears (Tamjid, 2025). STF treated fabrics absorb more energy and delay penetration more than untested controls, as these experiments consistently show. In contrast, comparisons with honeycomb cores show different densification and plateau behaviors. Under low velocity impacts, a recent TPMS-based metamaterial showed 50% less peak acceleration, demonstrating the adaptive stiffening offered by the shear-thickening transition (Corvi & Collini, 2024).

Gap of Knowledge

There are clear benefits if STFs are combined with metamaterials. The way that STF interacts with metamaterial structure in dynamic events is not fully described by current research. Not much is known about how pathways of deformation change once the STF becomes rigid, or how the onset of shear thickening. Additionally, the lack of standardized experiments in this field hinders the comparisons being made and slows the advancement (Zhao, 2022). This research will improve the application of basic science and engineering by directly addressing these gaps. This will most likely clarify the mechanisms of STF and strain rates and set the general knowledge for future researchers by creating repeatable procedures for multi-axial testing.

Currently, studies have been restricted to uniaxial impact or penetration testing. The research in current literature shows that shear-thickening fluids (STFs) significantly increase

energy absorption in protective fabrics (Wei, Lin, & Sun, 2022). Recent research on STF-filled Miura-ori structures has only been demonstrated through improvements in crushing force and energy absorption at high velocities (Sun, Tse, Hameed, et al., 2024), and TPMS composites have increased energy absorption by up to 85% under low-velocity impacts (Corvi & Collini, 2023). Much research combining STF has only been through fabrics or origami shapes. Therefore, filling this gap is essential. However, the behavior of STFs under combined compression, shear, or torsion loading is unknown because all experiments were conducted under uniaxial impact or penetration conditions (Chen, 2024). While honeycomb lattices collapse and absorb energy, auxetic structures are found to endure deformation and distribute impact forces over larger areas. However, the majority of studies have tested these compressions in one direction, just like with the STFs themselves. Research on STF-filled Miura-ori structures demonstrates that the fluid's rheology works in concert to improve energy absorption and crushing force, particularly at greater impact velocities.

Initially, this would offer a uniform testing procedure, facilitating reliable comparisons among various fluid structure systems and expediting the creation of universally superior protective elements (Wei et al., 2022). By gaining this understanding, STFs would be seen as active, adjustable components in metamaterials rather than passive fillers. In order to develop novel methods for impact-mitigation in applications such as personal protection, automotive, and aerospace, scientists would start to consider fluid-based smart behavior as an essential design variable rather than an auxiliary coating (Liu, Fu, Cui, et al., 2022). For this research, the mechanical properties and compression behavior of 3D printed lattice structures filled with shear thickening fluids will be measured in dependent and independent variables. For example, the lattice structure, STF formulation of the particle concentrations, and strain rate during

mechanical testing are all examples of how independent variables would be measured (Alexe, 2024). The dependent variables like the quantitative will measure the energy is absorbed, elasticity, and deformation rate.

Scientific Community

Metamaterials with STF systems can have a direct benefit to real world applications. The significance of this research extends beyond social application and scientific comprehension. The project will fill major scientific gaps in the understanding of fluid structure interaction under biaxial loading. The work will determine how rheological transitions function by altering STF formulations and metamaterial shapes. These results provide information that can be used to validate computation models and direct future experimental investigations in research domains like experimental mechanics, materials science, and rheology (Żurowski, 2021).

The project develops standard methods for testing STF metamaterial components under biaxial load in addition to advancing theory. This study makes it easier to create a foundation upon which scientific researchers can build by presenting repeatable procedures and thorough data (Mizzi, 2023). The methodological contribution will impact the direction of future research in the field and be valuable.

Public View

The potential advantages are immediate and apparent for society overall. To increase protection in athletic gear, such as preventing concussions or bone damage, STF composites can be used in protective devices. The addition of a new lightweight, flexible body armor that can absorb the shock caused by a sudden impact, can benefit the police and military. STF materials can be used to create impact shields such as bullet proof vests and crash absorbers for the automotive vehicles and aerospace (Tamjid, 2025). Adaptive protective materials provide

significant improvements even in everyday consumer goods like industrial protective gear and phone covers. Additionally protection gears may not need to be replaced frequently due to increased performance and durability, which would reduce long term expenses. Users could also be more obligated and use protective gear more frequently if it is more comfortable and effective (Huo, 2020). In addition to advancing scientific understanding, the study addresses important human needs for safety and adaptability in dangerous situations.

Methodology

Shear Thickening Fluid

First, the industrial grade fumed silica powder was dispersed into polyethylene glycol 400 (PEG 400) to create the shear thickening fluid mixture. Specifically, silica powder is an amorphous solid commonly used as a thickening agent due to its high surface area, while PEG 400 is a low molecular weight polymer that serves as a stable and effective carrier fluid (Huo, 2020). This combination allowed the STF to remain fluid-like under low stress while exhibiting shear-thickening behavior under high strain rates. Four different STF samples with varying silica concentrations in order to observe the general effect of particle concentration on viscosity and mechanical response. Keeping the PEG 400 at a constant variable, the industrial grade fumed silica were added based on the fluidity of the mixture. As silica was the thickening agent, more grams of silica compared to the 1ml of PEG400 created a dense viscosity. The silica-to-PEG ratios were calculated using weight percent measurements. For each sample, the silica mass fraction was converted into grams of silica and milliliters of PEG using the density of PEG 400.

Figure 1

STF Mixture

STF Type	Silica (g)	PEG 400 (ml)	Ratio	Evaluation
Low	0.126g	1ml	10 wt.%	Fluid
Average	0.484g	1ml	30 wt.%	Average viscosity
High	1.13g	1ml	50 wt.%	Dense viscosity
Control	0g	0ml	0 wt.%	Newtonian fluid

Note. Ratio of silica in grams to PEG in ml (Serra, 2024)

3D Lattice Structure

The lattice structure was modeled on Onshape seen in Figure 2. The data was collected using the same methods of lattice specimens printed on the 3D printed thermoplastic polyurethane with cavities made to hold the STF formulations (Sun, 2025).

Figure 2

Digital Lattice Structure Model

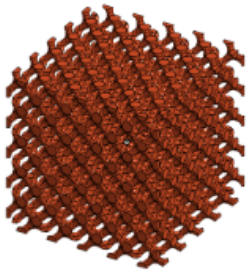


Figure 3

Digital (LS) Model Code

```
var Xcenter = (bbx1 - definition.HexWidth) / meter;
var Ycenter = (bby1 + (y * (Yincrement * meter))) / meter;

if (y % 2 != 0) // odd
    Xcenter += (FullWidthDimless / 2);

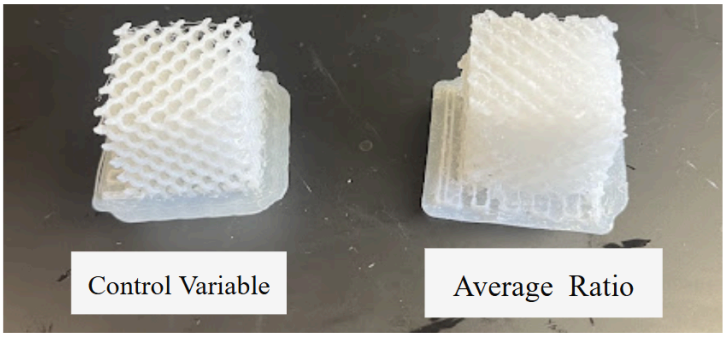
for (x = 0; x < TotalXcells; x += 1)
```

Figure 3, demonstrates the code utilized to repeat each layer of the lattice structure being designed. Specifically, every old row the code shifts the pattern half a cell width. These molds now have empty pockets that could be filled with Newtonian fluids with corresponding viscosities. This design guaranteed uniform distribution when the shear thickening fluid formulations were infused into the lattices' cavities. During the fabrication, TPU filament 95A was used for a stiffer rigid like structure. Thermoplastic polyurethane is a type of plastic commonly used as a filament for 3D printing that creates a flexible product similar to rubber

(Darnal, 2023). In this study, TPU was selected because it is both flexible and durable. When tested on the Universal Testing Machine, the flexibility of TPU allowed it to deform under load without fracturing. Common 3D printing filaments such as PLA have a hard structure, it is impossible to deform without a breakage point. This property is particularly important for this study, as it allows for more accurate observation of how the lattice behaves when infused with a shear-thickening fluid. The flexibility of TPU closely mimics conditions where STFs are most effective, enabling clearer analysis of changes in stiffness and energy absorption.

Figure 4

STF Treated Lattice Structure vs. Untreated Lattice Structure



Note. The left model is the control without STF, right contains STF.

In order to create control specimens, the cavities were either left empty. The experimental variables were filled with Newtonian fluids that have different viscosities comparable to the STF carrier liquid. In Figure 4, the lattice model on the left serves as the control specimen and was not infused with any STF. The model on the right was infused with the STF mixture prepared earlier. Comparing these two samples allowed direct observation of the mechanical influence of the shear-thickening fluid.

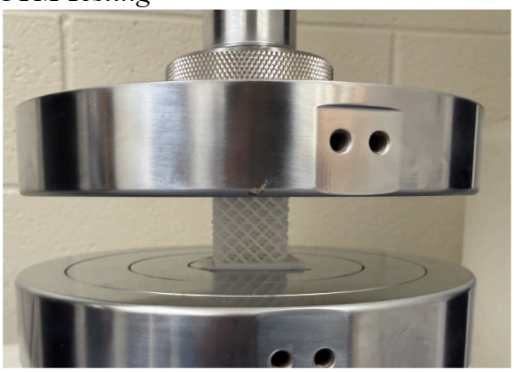
Variables

The UTM was then used to apply controlled compressive forces to the specimen. This created a graph that records the force applied to the lattice against the deformation, to help

analyze properties such as stiffness, energy absorption, and the effect of the STF on the lattice's mechanical performance.

Figure 5

UTM Testing



Note. Lattice specimen mounted in the universal testing machine

Figure 5 presents load vs extension testing. The independent variable was the applied load from the UTM. The dependent variable was the extension or deformation of the lattice structure, which was measured as the lattice stretches or compresses in response to the applied load. This is what is influenced as a result of the independent variable and reflects the mechanical behavior of the lattice, to observe how the STF contributes to increased resistance to deformation and improved energy dissipation. As current research states the lack of multiaxial loading, these specimens were tested in cyclic compressions. Cyclic compression on the UTM applied repeated, pulsed compressive loads to the material that measured its fatigue behavior, durability, and deformation over time rather than just its breaking point (Oni, 2025). This helped simulate real-world situations where protective appliances would have to withstand multiple compressions.

For this research, the mechanical properties and compression behavior of 3D printed lattice structures filled with shear thickening fluids will be measured in dependent and independent variables. For example, the lattice structure, STF formulation of the particle

concentrations, and strain rate during mechanical testing are all examples of how independent variables would be measured (Alexe, 2024). The dependent variables like the quantitative measures such as how the energy is absorbed, elasticity, and deformation rate.

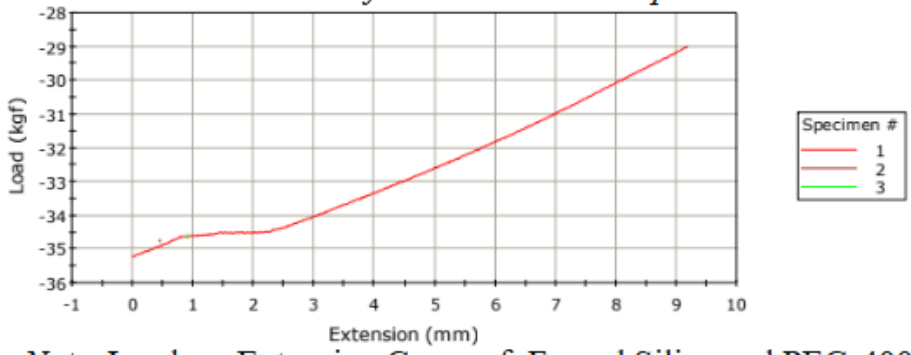
Results

Data Analysis

Based on the data collected, The curve shows stable, and mostly linear compressive behavior after the initial seating, or settling, of the material under load. When the Lattice structure is infused with the STF, the mixture makes the model more stabilized. This effectively reduces the major buckling effect. In FIgure 6, the curve does not drop at the end of the test, which suggests that the STF-infused lattice increased the specimen's load capacity. This indicates that the shear thickening fluid enhances the mechanical performance of the lattice by resisting deformation and improving energy absorption.

Figure 6

Load vs. Extension Curve for 30 wt.% *Lattice Specimen*



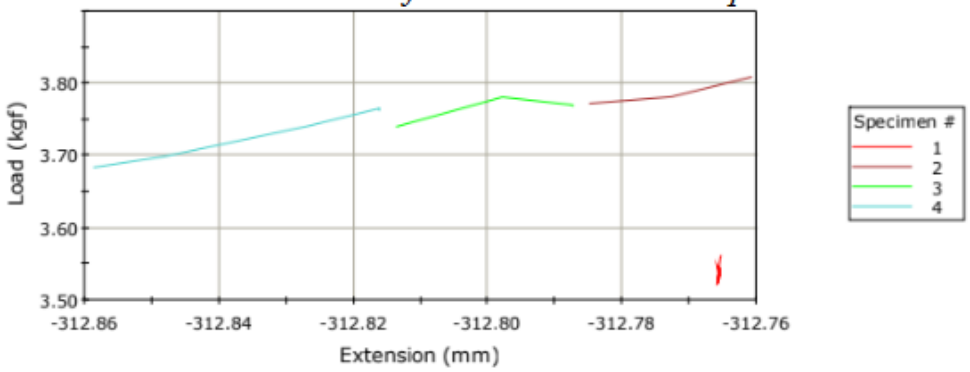
Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Figure 6 shows the data for the average viscosity STF mixture with a concentration of 30wt.%. Based on this graph, the integration of STF enhanced the lattice's resistance to deformation and improved the stability. After the initial bulking near 2mm, the slope of the graph

shows a smooth increasing linear behavior. The compressive response becomes smooth and nearly linear after initial seating,the load–deformation curve rises steadily (from approximately −34.7 kgf to −28.5 kgf) which suggests that these structures enhance structural response. Additionally, there is an absence of load drop near the end of the compression which suggests that the STF infused lattice structure is able to maintain its structure through the entire deformation. Overall, combining STF with metamaterial lattice structures enhances energy absorption and mechanical performance under realistic loading conditions.

Figure 7

Load vs. Extension Curve for ALL wt.% Lattice Specimens

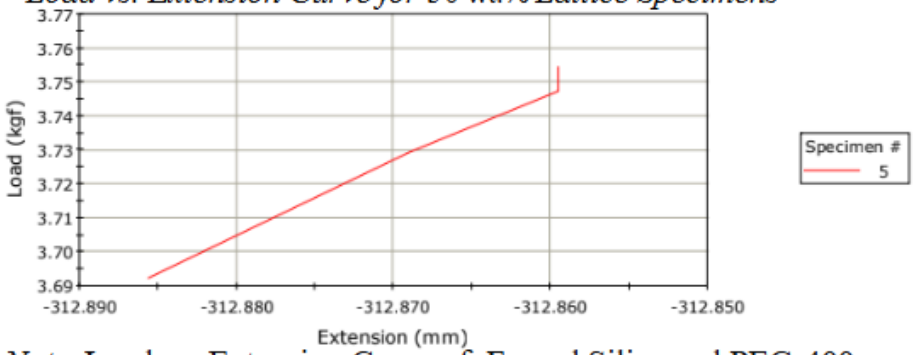


Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Figure 7 compares the performance of the control, 10 wt.%, and 30 wt.% STF-infused lattice specimens. Specimen 4, the control variable, shows a linear load vs. extension. In contrast, the STF-treated specimens (10 wt.% and 30 wt.%) had distinct deformation points as the applied load increased. These irregularities in the curve suggest that the viscosity of these weight percentages within the lattice, could not withstand the pressure on the geometry under stress. While the STF enhanced resistance to initial deformation, the 10 wt.% and 30 wt.% STF mixtures had points where the structure was not the most optimal combination.

Figure 8

Load vs. Extension Curve for 50 wt.% Lattice Specimens



Note. Load vs. Extension Curve of Fumed Silica and PEG-400 Infused Lattice Structure

Figure 8 highlights the performance of the 50 wt.% STF mixture, where the ratio of silica to PEG 400 was the most optimal sample tested. This specimen exhibited the steepest slope in the graph, which showed the highest stiffness and greatest resistance to deformation. The increased concentration of silica particles likely enhanced the shear thickening effect. Figure 8 showed an increase in viscosity under load.

Limitations

As seen in Figure 4, the lattice structures infused with varying shear thickening fluid mixtures were not sealed, which caused an experimental limitation. While testing in the universal testing machine, the pressure of continuous compressive forces caused the STF to progressively leave out of the internal cavities of the lattice. This leakage was because of the open cell geometry of the structure which did not have sufficient containment for the fluid under load. While there wasn't significant amounts of fluid squeezing out of the lattice structure, this behavior of STF under continuous loading may have had several effects on the experimental results. First, it may have limited the effectiveness of STF within the structure over time. The material being tested potentially caused inconsistent data throughout the duration of the trial.

Since the mechanical response of STF infused structures depended heavily on the distribution of the fluid, this likely led to an underestimation of the true energy.

Implications

After further research on potential seals for the STF-infused variables, elastomers were a potential solution. This is because elastomers are known for their rubber-like elasticity that returns to their original shape (Huo, 2020).

Figure 9

Elastomeric epoxy

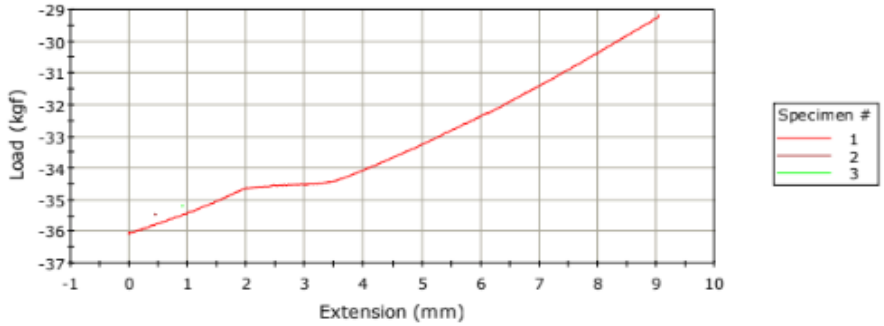


Applying this over the lattice structure for it to seal created a barrier for the STF. This containment is essential for real-world applications, as it ensures user safety by preventing exposure to the fluid while maintaining the protective performance of the material.

This new addition of adding a protective seal was then tested in comparison under the UTM. Using the same average density viscosity of 30wt.% from Figure 6, it created a direct comparison for data.

Figure 10

Load vs. Extension Curve Coated with Elastomeric epoxy



This additional data collection on UTM showed almost the same effects of an increased buckling behavior, but after this initial seating, it showed the same deformation curve.

Additionally while 3D printing the lattice structure, there were minor issues with the nozzle of the printer. As TPU is a flexible filament, the nozzle kept clogging up as the geometry had fine, intricate details. To solve this, the printing time setting was increased so the 3D printer could take each layer step by step. However this led to another issue where the filament dried too quickly causing the lattice geometry to have breakages in between.

Figure 11

Fabrication

Set of printing parameters for TPU

Layer Height [mm]	0.15
Nozzle Temperature [°C]	235
Infill [%]	100
Speed [mm/min]	45

To solve this limitation, the temperature of the printer was increased, as shown in Figure 11. As the temperature was increased from 150°C to 235°C, the nozzle temperature helped maintain the TPU filament in a more fluid state during printing. This solution enhanced the bonding between layers for the overall structure. This adjustment reduced the number of breakages and enhanced the durability of the model.

Discussions

Findings

The results of this research suggest that the integration of STF into metamaterial lattice structure has a positive impact on the mechanical performance under complex loading. Across all of the testing specimens, the 50wt.% ratio showed the greatest resistance to deformation. One of the most significant findings through this study was the reduction of bulking behavior with the integration of shear thickening fluids. The STF stabilized the lattice during the loading most evidently shown in the 30 wt.% and 50wt.% where the load vs. extension curves had the most smooth and linear slopes compared to the control variable.

The performance of different STF concentrations are compared and found to depend on the composition. The lower concentrations (10 wt.%) showed some improvement over the control but also showed localized points of deformation under sustained load. Higher concentrations , especially the 50 wt.% mixture , showed the steepest slope of load – extension , which means higher stiffness and more resistance to compressive forces . This suggests that the silica content increases the shear thickening effect, but more testing would be needed to find the optimal concentration range.

Overall, this finding supports the experimental hypothesis (H_1) that the integration STF with metamaterial lattice structures significantly enhances energy absorption and mechanical performance under complex loading conditions compared to metamaterial structures without STFs. This result adds onto the gap in the literature and suggests potential applications in impact resistant materials.

Ethics

This research was feasible with the conditions and resources available. A universal testing machine allowed controlled loading for biaxial compressions on STFs, while having access to 3D printers allowed for complex metamaterial structures to be created. STF preparation only required standard laboratory equipment for particle dispersion and mixing. Following the safety procedures would allow an appropriate experiment to be done. Additionally, neither biological samples nor humans are needed for this study. Working with STF ingredients like silica particles and carrier fluids was done with caution under a laboratory hood and could be done in the future (Żurowski, 2021). To avoid exposure, protective equipment like lab coats, gloves, and goggles were utilized. All items, such as fumed silica and polyethylene, as instructed by the manufacturer, were packaged for disposal and disposed of through a licensed chemical waste disposal. Additionally, thermoplastic polyurethane was mechanically recycled by grinding them into smaller particles for reprocessing. These steps could be taken for future applications as well.

Conclusion

The combination of metamaterials and shear thickening fluids are an innovative approach to the development of secure and protective materials. This study addresses the difference between multi axial forces of practical impacts. Additionally model studies by systematically evaluating STF embedded lattices. Both quantitative data and mechanistic insight were produced by the experiment, which combined qualitative research with quantitative mechanical testing. The research improves both scientific understanding and useful engineering design applications for materials by developing standardized testing data and quantitatively measuring fluid and structure interactions. Based on the result of slope in Figure 8, the 50wt.% had the optimal

amount of energy absorption. The combination of 1.13g of fumed silica powder to 1ml of PEG400. With more knowledge on the potential shear thickening fluids can be applied to in real life applications. Through the control of lattice structures such as auxetic, strain rate, and STF particle loading (low, medium, and high), as well as the inclusion of suitable controls, the study will produce an experimental matrix that can isolate main effects and interactions (Chen, 2024).

Future Study

Future work should focus on expanding both the material and testing conditions to further enhance the performance of shear thickening fluid infused lattice structures. First, a wider range of silica-to-PEG ratios should be tested for a wider range to collect data on how composition influences mechanical properties such as different stiffness, energy absorption, and deformation behavior. Additionally, exploring alternative materials such as polyvinyl alcohol (PVA) gels. PVA gels are known to have similar properties such as PEG 400. However, PVA gels are known to be more environmentally friendly while maintaining the same mechanical characteristics. The temperature dependence of STF should also be further studied. When applied to real-world changes in environmental conditions can affect viscosity and shear response.

On the other hand, to strengthen the reliability of the results, additional data should be collected using the universal testing machine. Collect additional data using the UTM under standard compression, ballistic loading, and creep testing for more quantitative data on the effectiveness of STF infusion. Future steps should examine potential degradation mechanisms such as particle settling or phase separation over time. Additionally, future research should address scalability and manufacturability for real-world applications. For example, cost calculations, material efficiency, and the feasibility of large production methods should be taken into account.

References

Alexe, F., Sau, C., Iorga, O., Toader, G., Diacon, A., Rusen, E., Lazaroaie, C., Ginghina, R. E., Tiganescu, T. V., Teodorescu, M., & Sobetkii, A. (2024). Experimental Investigations on Shear Thickening Fluids as “Liquid Body Armors”: Non-Conventional Formulations for Ballistic Protection. *Polymers*, 16(16), 2305. https://doi.org/10.3390/polym16162305

Bajya, M., Majumdar, A., Butola, B. S., Verma, S. K., & Bhattacharjee, D. (2020). Design strategy for optimising weight and ballistic performance of soft body armour reinforced with shear thickening fluid. *Composites Part B: Engineering*, 183, 107721. https://doi.org/10.1016/j.compositesb.2019.107721

Chen, Z., Jing, Y., Wang, H., Zhang, X., Long, S., & Yao, X. (2024). Fluid-structure interaction mechanism in shear thickening fluid/Foam composite for advanced protective structures. *Materials Today Communications,* 41, 110662. https://doi.org/10.1016/j.mtcomm.2024.110662

Corvi, A., & Collini, L. (2024). Shear-thickening-fluid-based meta-material for adaptive impact response. *Materials & Design*, 244, 113174. https://doi.org/10.1016/j.matdes.2024.113174

Darnal, A., Shahid, Z., Deshpande, H., Kim, J., & Muliana, A. (2023). Tuning mechanical properties of 3D printed composites with PLA:TPU programmable filaments. *Composite Structures*, 318, 117075. https://doi.org/10.1016/j.compstruct.2023.117075

De la Rosa, S., Mayuet, P. F., Méndez Salgueiro, J. R., & Rodríguez-Parada, L. (2021). Design of Customized TPU Lattice Structures for Additive Manufacturing: *Influence on the Functional Properties in Elastic Products.* Polymers, 13(24), 4341. https://doi.org/10.3390/polym13244341

Gorguluarslan, R. M., Gandhi, U. N., Mandapati, R., & Choi, S. K. (2015). Design and fabrication of periodic lattice-based cellular structures. *Computer-Aided Design and Applications,* 13(1), 50–62. https://doi.org/10.1080/16864360.2015.1059194

Huo, J.-L., Sun, F., Li, T.-T., Shiu, B.-C., Lou, C.-W., & Lin, J.-H. (2020). Preparation and properties of shear thickening fluid (STF) capsule filled graded buffer composites. *Journal of Materials Research and Technology*, 9(5), 10982–10990. https://doi.org/10.1016/j.jmrt.2020.07.106

Hou, S., Li, T., Jia, Z., & Wang, L. (2018). Mechanical properties of sandwich composites with 3D-printed auxetic and non-auxetic lattice cores under low velocity impact. *Materials & Design*, 160, 1305–1321. https://doi.org/10.1016/j.matdes.2018.11.002

Jiang, H. (2023). Engineering 3D architected metamaterials for enhanced mechanical properties and functionalities (Doctoral dissertation). *Electronic Theses and Dissertations*, Paper 4053. https://doi.org/10.18297/etd/4053

Lee, H., Eom, R.-i., & Lee, Y. (2019). Evaluation of the mechanical properties of porous thermoplastic polyurethane obtained by 3D printing for protective gear. *Advances in Materials Science and Engineering*, 2019, Article 5838361. https://doi.org/10.1155/2019/5838361

Lee, W., & Laufer, F. (2019, September 9). Development of a Desktop Universal Testing Machine with a Full-field Optical Extensometer. *Sensors and materials*. https://doi.org/10.18494/SAM.2019.2368

Liu, H., Fu, K., Cui, X., Zhu, H., & Yang, B. (2023). Shear Thickening Fluid and Its Application in Impact Protection: *A Review. Polymers,* 15(10), 2238. https://doi.org/10.3390/polym15102238

Mizzi, L., Hoseini, S. F., Formighieri , M., & Spaggiari, A. (2023, February 10). Development and prototyping of SMA-metamaterial biaxial composite actuators. *IOP Science.* https://iopscience.iop.org/article/10.1088/1361-665X/acb749

Oni, S. B. (2025, April 27). A Statistical Approach to Data Reduction in UTM-LST External Balance Assessment. *Research Gate.* https://www.researchgate.net/publication/391318498_Development_of_a_Real-Time_Data_Reduction_Algorithm_for_UTM-LST_External_Balance_Applications

Sun, H., Tse, K. M., Hameed, N., & Lu, G. (2025). Dynamic compressive behavior of Miura-ori metamaterials filled with shear thickening fluid. *International Journal of Mechanical Sciences*, 288, 110006. https://doi.org/10.1016/j.ijmecsci.2025.110006

Tamjid, M. I., Abtew, M. A., & Kopot, C. (2025). Lightweight Textile and Fiber-Reinforced Composites for Soft Body Armor (SBA): Advances in Panel Design, Materials, and Testing Standards. *Journal of Composites Science*, 9(7), 337. https://doi.org/10.3390/jcs9070337

Wei, M., Lin, K., & Sun, L. (2022). Shear thickening fluids and their applications. *Materials & Design*, 216, 110570. https://doi.org/10.1016/j.matdes.2022.110570

Yuan, S., Chua, C. K., & Zhou, K. (2018, December). 3d‐printed mechanical metamaterials with high. *Wiley Vch*. https://www.researchgate.net/publication/329406974_3D-Printed_Mechanical_Metamaterials_with_High_Energy_Absorption

Zhao, F., Wu, L., Lu, Z., Lin, J.-H., & Jiang, Q. (2022). Design of shear thickening fluid/polyurethane foam skeleton sandwich composite based on non-Newtonian

fluid–solid interaction under low-velocity impact. *Materials & Design,* 213, 110375. https://doi.org/10.1016/j.matdes.2021.110375

Żurowski, R., Falkowski, P., Zygmuntowicz, J., & Szafran, M. (2021). Rheological and Technological Aspects in Designing the Properties of Shear Thickening Fluids. *Materials,* 14(21), 6585. https://doi.org/10.3390/ma14216585